MIL-L-28583C(YD)
5 September 1986
SUPERSEDING
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28 January 1979

MILITARY SPECIFICATION

LIGHT, BED: FLEXIBLE ARM, HEAVY DUTY

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements for materials and their assembly into a sturdily constructed adjustable light.

1.2 Classification. Lights shall be of two styles, as specified (see 3.1 and 6.2.1).

Style A - Clamp-on style.

Style B - Bolt-on style.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A | FSC 6230

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SPECIFICATION

MILITARY

MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Provisioned Items (Repair Parts): Packaging of.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized Unit Loads.

(Copies of specifications and standards, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS shall be the issue of the non Government documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D3881 - Phenolic Transfer and Injection Molding Compounds.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL BUREAU OF STANDARDS (NBS)

Handbook H28 - Screw-Thread Standards for Federal Service.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

UNDERWRITERS LABORATORIES INC. (UL)

UL 62 - Flexible Cord and Fixture Wire.

(Application for copies should be addressed to the Underwriters Laboratories Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The lights shall be as shown on figure 1 and as described below. The lights shall be adjustable, operate on 110 volt (V), 60 cycle alternating current (ac), and use a medium screw base 60 watt (W) bulb. The outer shade shall be plastic and the inner reflector shall be aluminum.

3.1.1 Light, style A, clamp-on style. The style A light shall be capable of being clamped to a minimum 5/8-inch thick panel and a maximum 1-1/2-inch diameter tube on bunk bed corner post. The light adjustment shall be accomplished by having the capability of being raised and lowered by the coiled flexible arm and having a limited rotational movement of the light reflector or head.

3.1.2 Light, style B, bolt-on style. The style B light is for permanent installation on a wood bunk bed panel or bookshelf. The light adjustment shall be accomplished as in 3.1.1. Carriage bolts and pallet nuts which require a 1/4-inch allen wrench for installation and removal are specified for this style of light to deter removal by unauthorized personnel.

3.2 First article. When specified (see 6.2.1), the contractor shall furnish a light for first article inspection (see 4.3.1 and 6.4).

3.3 Standard commercial product. The light shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the light being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.4 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.5 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means

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materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.5.1 Electrical socket. The electrical socket shall be white glazed porcelain for use with medium base bulb and rated for a minimum of 660W and 250V. The socket shall be provided with a tapped cast cap for fastening to the back of the outer shade. The light cord exit shall be at the socket side, through the porcelain wall.

3.5.2 Light cord. The light cord shall be in accordance with UL 62, type SPT-1. The usable exposed cord shall be 6, 8, 10, or 12 feet long, as specified (see 6.2.1). A line switch shall be provided on the cord 12 inches, ± 1/4 inch, from the flexible arm.

3.5.3 Reflector. The light reflector shall consist of an outer shade and inner reflector.

3.5.3.1 Outer shade. The plastic outer shade shall be black phenolic plastic conforming to ASTM D3881. The form, thickness, and dimensions shall be as shown on figure 2 or figure 8.

3.5.3.2 Inner reflector. The inner reflector shall be a minimum 0.040-inch thick spun aluminum of the form and dimensions shown on figure 3. For fastening the inner reflector to the back of the outer shade, a steel strap shall be secured to the back end of the inner reflector by not less than two rivets at each end. Four equally spaced stamped protrusions, as shown on figure 3, shall be provided at the front of the inner reflector to keep the spacing equal between the outer shade and the inner reflector. Finish shall be as specified in 3.7.1.

3.5.3.3 Warning sticker. An adhesive-backed paper sticker shall be placed on the inside surface of each inner reflector. The sticker shall provide a warning that the light bulb to be used does not exceed 60W.

3.5.4 Flexible arm. The coiled flexible arm shall be 15 inches long overall, ± 1/4 inch, stiff or industrial grade with an outside diameter of 0.610 inch, ± 0.010 inch. Each end of the arm shall be provided with a nonturnable crimped-on threaded steel coupling. Finish shall be as specified in 3.7.2.

3.5.5 Clamp. For style A lights, the clamp shall be a minimum 0.090-inch thick steel except threaded members. Dimensions and forms shall be as shown on figure 4. A 1/4-20 or 5/16-18 thread, 1-1/4-inch long thumb screw shall operate the clamp movable member through a clinch nut brazed or silver soldered to the clamp. The thumb screw shall be pinned or staked fast to the clamp movable member. Clamp interior surfaces shall be covered with non-slip type pads as shown on figure 4. Pads shall be cork-neoprene type 1/32-inch thick, adhesive backed, similar and equal to that manufactured by Mechanical Felt and Textile Co. A 7/8-inch minimum length coupling with a side outlet for exit of the light cord shall be positioned on the clamp as shown on figure 4 and brazed or silver soldered to the clamp. Brazing or soldering shall be

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around the entire perimeter of the mating surfaces. The letters "US" denoting government property, shall be stamped or embossed in characters 3/8 inch high on the clamp exterior as shown on figure 4. Finish shall be as specified in 3.7.3.

3.5.6 Bolt-on fastener. The bolt-on fastener shown on figure 5 shall be 0.090-inch thick steel. Unless otherwise specified (see 6.2.1 and 6.5), two 1/4-20 thread, 1/2-inch long carriage bolts and two 1/4-20 thread pallet nuts shall be provided with each fastener. The square holes in the fastener shall be a close fit to the carriage bolt shank to prevent rotation of the carriage bolt. A 7/8-inch minimum length coupling shall be positioned on the bolt-on fastener as shown on figure 1 and brazed or silver soldered around the entire perimeter of the mating surfaces. The fastener shall be marked for identification at location shown on figure 5 in same manner as specified in 3.5.5.

3.6 Light assembly. The light cord shall exit through a plastic bushing as shown on figure 1. The cord shall be provided with a strain relief. The exposed cord inside the reflector, including the strain relief knot, shall be protected with full length fiberglass sleeves. Light cord shall be fastened to socket terminals by soldering or screws. The outer shade shall be fastened to the flexible arm crimped-on coupling by a lug adapter and further secured with a liquid compound which solidifies into a permanent set. Two bow washers shall be installed between the interior surface of the outer shade and the lug adapter to allow the reflector to be turned 345 degree (°) $\pm$ 5°. A tab on the lug adapter shall act as a stop to prevent full rotation of the reflector. The socket with a tapped cast cap and the steel strapped aluminum reflector, figure 3, shall be fastened to the outer shade with a truss-slotted head screw. As an alternate, a nickel plated 1/8-27 hex head cap nut and 1/8-27 by 5/8-inch milled nipple may be used for fastening the socket to the outer shade. The flexible arm threaded crimped-on coupling shall be fastened to the clamp being threaded into the coupling of the clamp or bolt-on fastener as specified. Assembled lights shall pass the tests specified in 4.6.

3.7 Finish. Finishes shall be smooth, without dirt, dust, or other foreign matter embedded and shall not be discolored, rippled, peeled, have sags or runs, or bare spots.

3.7.1 Aluminum inner reflector. Finish shall be bright anodized.

3.7.2 Flexible arm. Finish shall be electroplated brass or chromium, as specified (see 6.2.1).

3.7.3 Clamp and bolt-on fastener. All components, except the threaded members, shall be finished with a black enamel over a phosphate coating.

3.7.4 Hardware components. Thumbscrews, pallet nuts, carriage bolts, inner reflector steel strap, and unexposed hardware components shall be chromium plate or similarly treated for corrosion resistance.

3.8 Workmanship. Workmanship shall be of the highest quality throughout in accordance with the best standard practices used in the manufacture of electrical lights of this type.

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3.8.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.8.2 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.8.3 Mechanical fasteners. Holes for securing components by mechanical fasteners such as screws and bolts shall be accurately punched or drilled and shall have the burrs removed. All bolts, nuts, and screws shall be tight. All nuts, bolts, and screws shall have screw threads in accordance with NBS Handbook H28.

3.9 Light installation and adjustment. Figure 6 illustrates adjustment instructions for style A lights and installation instructions for style B lights. A copy of figure 6 or similar instruction is to be furnished with each packaged light in accordance with customary trade practice.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on each light when a first article is required (see 3.2 and 6.2.1). This inspection shall include the examination of 4.4, the tests of 4.5, and, when specified, the first article pack inspection of 4.6 (see 4.6 and 6.2.1). The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.4, the tests of 4.5.1, 4.5.2, and 4.5.3 and the packaging inspection of 4.6. This inspection shall be performed on the samples selected in accordance with 4.3.

4.3 Sampling. Sampling and inspection procedures shall be in accordance with MIL-STD-105. The unit of product shall be one light. All lights offered for delivery at one time shall be considered a lot for the purpose of inspection. The inspection level shall be level II and the Acceptable Quality Level shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for a complete reinspection. Resubmitted lots shall be reinspected using tightened inspection. If the rejected lot was screened, reinspection shall be limited to the defect causing rejection. If the lot was reprocessed, reinspection shall be performed for all defects. Rejected lots shall be separated from new lots, and shall be clearly identified as reinspected lots.

4.4 Examination. Each light shall be examined for defects listed in table I. Each attribute within each classification of multiple defects shall constitute one defect.

4.5 Tests.

TABLE I. Classification of defects.

Classification	Defects	Requirement paragraph
Critical	None defined.	
Major:		
101	Light not style specified.	3.1
102	Design not as specified.	Figure 1
103	Adjustment capabilities not as specified.	3.1.1 & 3.1.2
104	Electrical socket not as specified.	3.5.1

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TABLE I. Classification of defects. cont'd

Classification	Defects	Requirement paragraph
105	Electric socket cord exit not as specified.	3.5.1
106	Light cord not type SPT-1.	3.5.2
107	Light cord not of length specified.	3.5.2
108	Outer shade not as specified.	3.4.3.1
109	Inner reflector not as specified.	3.5.3.2
110	Flexible arm not length specified.	3.5.4
111	Flexible arm not of diameter specified.	3.5.4
112	Clamp steel not of thickness specified.	3.5.5
113	Thumb screw not of length and size specified.	3.5.5
114	Clamp pads missing or not of material specified.	3.5.5
115	Coupling not secured to the clamp as specified.	3.5.5
116	When bolt-on fastener is required, not as specified.	3.5.5
117	Light cord exit through coupling not provided with a bushing.	3.6
118	Strain relief knot not protected by fiberglass sleeve.	3.6
119	Outer shade not secured as specified.	3.6
120	Turn of reflector not as specified.	3.6
121	Finish for various components not as specified.	3.7
122	Workmanship not as specified	3.8
Minor:		
201	Location of line switch not as specified.	3.5.2
202	Warning sticker missing or not provided.	3.5.3.3
203	Identification marking missing or not as specified.	3.5.5
204	Light installation and adjustment instructions not provided.	3.9

4.5.1 Flexible arm deflection test. As shown on figure 7, secure the light clamp in a vise or similar holding fixture in a manner to permit the flexible arm and reflector to be extended horizontally, parallel to the floor, and free of any obstructions. Test the stiffness of the flexible arm by hanging a weight totaling 200 grams on the flexible arm at the point nearest the reflector. The downward deflection of the reflector shall not exceed 1/2 inch. Further deflection will constitute failure of this test.

4.5.2 Dielectric strength. The light shall withstand, without breakdown, a potential of 900V, 60 Hertz (Hz), ac applied between current carrying parts, and noncurrent-carrying metal parts and between current-carrying metal parts

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and between current-carrying parts of opposite polarity of a period of one minute. During this test, the equipment shall be at a temperature of 70° Fahrenheit (F) to 80°F and a relative humidity of from 50 to 70 percent.

4.5.3 Leakage current. The sample light shall be connected to a 115V, 60 Hz ac supply circuit. The leakage current measured between the ungrounded metal frame of the lamp and each of the supply terminals shall not exceed 0.2 milliampere under the following conditions:

a. Initially at a room temperature and humidity of not less than 70°F (21° Centrigrade (C)) and 50 percent relative humidity. If a line switch is included in the light, this measurement shall be made both before and after the switch is closed.
b. After exposure for 24 hours in an atmosphere of not less than 85 percent relative humidity and temperature of 85°F ± 5°F (29° ± 3°C), and while still under these conditions, if a line switch is included in the light, this measurement shall be made both before and after the switch is closed.

4.6 Packaging inspection. The inspection of the preservation, packing, and marking shall be in accordance with the requirements of section 4 of MIL-E-17555. The inspection shall consist of the quality conformance inspection; and, when specified (see 6.2.1), a first article pack shall be furnished for examination and test within the time frame required (see 6.2.1).

5. PACKAGING

5.1 Preservation, packing, and marking. Preservation, packing, and marking shall be in accordance with the requirements of MIL-E-17555 with the level of preservation and the level of packing as specified (see 6.2.1).

5.2 Palletization. Material shall be palletized in accordance with MIL-STD-147 when the following criteria are met:

a. Load to consist of four or more unskidded containers; and,
b. Load shall utilize a minimum of 80 percent of the pallet base.

5.3 Marking. Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The lights are intended for use with military standard bunk beds as shown on figure 6.

6.2 Ordering data. Procurement documents should specify the following:

6.2.1 Procurement requirements.

a. Title, number, and date of this specification.
b. Style of light required (see 1.2).
c. When a first article is not required for inspection and approval (see 3.2, 4.2.1, and 6.4).
d. Length of light cord required (see 3.5.2).

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e. For style B, specify carriage bolt length if other than 1/2 inch (see 3.5.6 and 6.5).
f. Finish required for flexible arm (see 3.7.2).
g. When preservation, packing, and marking is required (see 4.6).
h. Level of preservation and packaging, and level of packing required (see 5.1).

6.3 Light bulb information. These lights are designed to use medium screw base, 60W bulbs. Due to higher shipping costs and breakage, bulbs are not included. Light bulbs are available through usual supply channels. Adequate number of bulbs, including replacements, should be ordered. Bulbs are available from General Services Administration, Federal Supply Schedule FSC Group 62, Part III. The National Stock Number for medium screw base, 60W bulb is NSN 6240-00-635-4480.

6.4 First article. When a first article inspection is required, the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.5 Installation of style B, bolt-on light. The bolt-on light is for permanent installation on wood bunk headboard panels or bookshelves not more than 7/8 inch thick. If the lights are to be mounted on panels thicker than 7/8 inch, carriage bolts longer than the 1/2-inch length specified are required.

6.7 Subject term (key word) listing.

Illuminating interior lighting
Lamps - incandescent

6.8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing Activity:

Navy - YD

(Project 6230-N306)

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